

Living as a Service (LaaS): An Experiment to Evaluate Living Experience in Connected Homes in the Netherlands

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ABSTRACT

User-centric approaches to technological interfaces have increasingly surfaced in the recent times. In smart homes, this manifests in the form of how people adapt and interact with devices, and the set up as a whole. This study aims to analyze the behavioral aspects of living in a smart home through interview and structured survey-based interactions with 21 families in the Netherlands who were provided a predefined set of smart devices for about three weeks. We evaluate user experience with smart devices through four stages of smart home adoption. Based on user behavior and motivations, we find four personas emerging, namely convenience-oriented, safety-oriented, cautiously supportive and sustainability-oriented. The study finds a positive experience predominantly with smart home devices when they are delivered and integrated in a manner that serves user's priorities in a home. Future directions incorporating user preferences in smart home design are discussed.

CCS CONCEPTS

• User characteristics; • Usability in security and Privacy;

KEYWORDS

Smart home, Behavioral factors, User journey, User persona

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1 INTRODUCTION

User centric approaches on technological interfaces have increasingly surfaced in the recent times. Today, users have become the

focal point in creation of services and design best suited to meet their requirements. With advances in IoT and cloud computing, there has been an uptick in the development of smart home applications. While smart devices per se are built on advances in digital technologies, the vital ingredient for the recipe to succeed would be to understand why individuals would need them, what they look for from these devices and how a setup of such devices can influence people's living for the better. In particular, it would be interesting to understand the behavioral underpinnings of smart home living, the motivations, the preferences and the perceptions. Our study carried out an in-situ experiment involving smart home dwellers in Netherlands for a duration of three weeks. We discuss the results of the study in terms of user experiences in the three phases of pre-installation, installation and post-installation of the smart devices in their residences and discuss user personas based on the user's behavioral attributes.

2 RELATED WORK

Understanding user needs and experiences can help in designing smart homes that are responsive customer needs. For instance, home automation could be beneficial for elderly people for emergency help, monitoring seniors' health etc. [13]. In another study [14], they tried understanding what the participants defined as "smart" and their motivations for home automation. A diary study conducted with 20 participants to assess IOT device sharing behavior looked at motivators of shared use influenced by user preferences and motivations. Similarly, aspects of coordination among devices, accommodating needs of multiple co-users, content sharing/analysis and trust among users were also looked at [15]. A useful way to understand and categorize user needs is the use of personas, which are detailed descriptions of a hypothetical user based on their goals and needs [16]. Personas may be based on a combination of real and fictitious data [17] and could be driven by qualitative data [16], quantitative data [18] or online analytics [17]. A study of smart home users' desires [19] finds three emergent personas based on qualitative data on user's utility from smart home, namely the helper who focuses on functional utility of a smart home, the optimizer who desires the fulfilment of expected outcomes and the hedonist who is inclined to the aesthetic and pleasure-giving aspects. One study [20] looks at the lifecycle of a smart home in a 3-week study involving eight families and presents a model of six stages of user experience ranging from installation to removal of devices. The stages identified were initial installation,

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Table 1: Operational definitions of key variables

Variable	Definition
Environment consciousness	Individual's expressed willingness to adopt measures like energy saving or to allocate resources for conservation.
Exploratory behavior	Those behaviors in which an individual appears to be taking a certain initiative in finding out more about their environment.
Technology Adoption	The usage behavior of individuals adopting the information technology.
Impulsivity	Readiness to take immediate and unplanned action as a response to internal and external stimuli, with no regard for their negative consequences for themselves or the others.
Saving behavior	The individual propensity to save in a given environment or context.
Social norms	Those expected forms of regular behavior whose absence or violation causes social sanctions.
Attitude	Disposition towards or against a specified phenomenon, person or thing.

motivation, implementation, use-through-routine (evaluation of the functioning of smart home applications), routinization (smart home applications becoming part of users' routine) and removal of devices. Our study also attempts to delineate stages in adaptation to smart devices and bring out the predominant motivations, emotions and concerns that emerge in each stage. We combine persona-based approach with depiction of stages of interaction with the devices in our study considering user dispositions towards adoption of smart homes in pre-experimental and experimental phases. We attempt to arrive at personas that are based on concrete requirements from a smart home so that service providers can make practical applications that serve these specific requirements.

3 METHODOLOGY

The smart home study was conducted in collaboration with a leading bank and an IT service provider who identified and recruited participants. The participants of the smart home experiment were employees of these two organizations who gave ready consent to be part of the study. After being given a predefined kit of smart devices, participants were requested to install it themselves and contact the service providers in case they needed any assistance. They were intimated in advance that an interview would be conducted during the experiment and questionnaire would be administered to them at the end. The smart home kit was entrusted with the participants for a minimum period of three weeks. The devices comprised of motion sensors, window sensor, door sensor, smart bulbs, smoke alarm, wireless camera, smart plug and dimmer switch. Once the smart home set up was completed and users had enough time to use and interact with the devices, there was a discussion with the participants, designed as semi-structured interviews. This initial interaction was aimed at understanding users' ideas of a smart home, their usual routine and experiences of installation and use of the devices. A survey was administered towards the end of the experiment that captured device-specific feedback and measures of behavioral parameters. The variables under study have been categorized into behavioral, experience-related, evaluation of utilities and concerns with the smart home setup. The behavioral parameters have been identified based on literature evidence in the space of technology adoption and usage. The key behavioral variables that have been analyzed in the study are given in Table 1. Responses on the experience of adapting to a smart home, general extent of

technology use, inclination towards technology and aspects like cost and privacy were also obtained. The device-specific parameters explored the number of devices installed, approximate time spent in installation, major utility derived from the device and ease in installation.

The connected homes experiment started in February 2019. We conducted twenty-one interviews. The average age of the participants was 36 (Min=25, Max=54, SD=7.33). 33% were nationals of India, while the rest were nationals of the European region. Almost all users had technology background either by training or work experience. 14% of the participants indicated that they lived alone.

4 STUDY INSIGHTS

4.1 User journey

We used an inductive approach in coding the user interview transcripts. First, we used codes corresponding to the experience of people over the lifecycle of engagement with the devices. This gave us four distinct stages, namely, Pre-experiment, Encountering the devices, Installation and Post-installation experiences, which are detailed out in this section (Figure 1).

4.1.1 Pre-experiment- Meaning of home and familiarity with smart devices. Responses to what a home meant to participants varied from 'a comfortable space to come back to and spend time with family' to 'a place to chill and relax after work' to 'just a place to sleep in'. While all participants had specific conceptualizations of what a home meant to them, two major dimensions stood out, namely comfort and security. While 76% of the participants emphasized the importance of comfort and freedom, 33% mentioned safety and security and 19% touched upon sustainability as priorities when they think of a home.

Those participants with children emphasized the importance of safety from burglaries and accidents, and convenience in terms of childcare being made easier with smart devices. Most of the participants expressed affiliation towards technology and devices in general and cited this as an important motivation for adopting the smart home setup.

4.1.2 Encountering the devices. The receipt of the setup kit was met with various emotions. While some were excited and set it up immediately, others were more apprehensive. The overwhelming emotions associated with setting up the devices were excitement

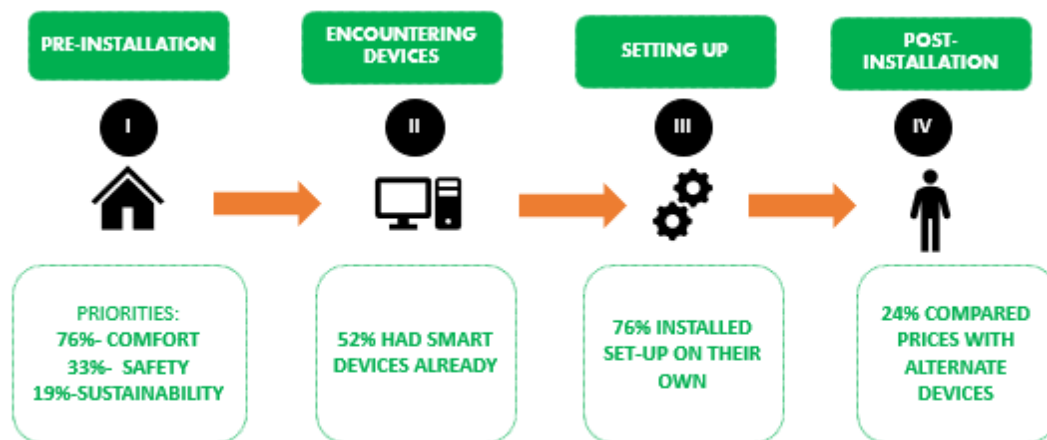


Figure 1: User journey with smart devices

and curiosity. Participants also mentioned that the kit made them more aware about smart homes although a few said they did not plan on investing in devices in the long term. 52% of the participants already had smart devices at home which included burglar alarm, thermostat, air quality checker and smart bulbs. The existing devices gave us some insight into their preferences. For instance, one of the participants who indicated an interest in energy saving and sustainability already had a device to detect energy use and expressed interest in buying a smart water meter to reduce water wastage. Another participant who was worried about burglary indicated plans to buy a web camera. A few participants had concerns such as why a device would be needed when switches were in place or were thrown off by its complexity.

4.1.3 Setting up. Participants expressed emotions ranging from having had fun to having ‘suffered a headache by the end of the process’. 76% participants installed the setup on their own while others took help from family members. Many were also keen on experimenting creatively with the devices, making unique connections such as setting the light to dim when TV is on. While some mentioned setting the rules on the app and requirement of a lot of configuration as the pain points, others mentioned issues such as devices not being recognized by the hub, the hub not being able to distinguish same devices of different brands and power shutting off for all devices when one is not at home. The devices that almost all participants set up or attempted to set up were motion sensors and smart bulbs. Some participants however were confused with motion sensors, unable to figure out the difference between motion sensor, door sensor and window sensor. Some mentioned that the motion sensor alerts irrelevant movements as well. Many participants faced issues with connecting smart bulbs to the hub but those who were able to set it up immensely enjoyed the ambience of their rooms with the new bulbs. While some users enjoyed the camera vastly and expressed interest to buy one for themselves after the experiment, others were worried about privacy issues and difficulty in setting it up. Some additional features users wished for in the setup included seasonal automatic time adjustment for

bulbs, reminders/notifications about device usage, compatibility with different socket sizes and additional user interface on the wall of the house.

4.1.4 Post-installation. The post-installation phase centered around adaptation to the smart home setup. Transition to a smart home wasn’t very seamless for some participants. They mentioned how it took some time to switch from their old habits to an automated system. However once accustomed to the devices, many had a positive reaction to the change in their home living experience. Overall, most users responded in the affirmative, when asked if they would recommend smart home living. Many described it to be futuristic and considered it to be an integral part of the modern living spaces. Many participants of the experiment cited the price of setting up the devices as a major limitation from using it. 24% of the participants mentioned how the devices could face strong market competition from cheaper products.

4.2 Behavioral Attributes of Smart home living

This section presents the findings from the questionnaire responses (Figure 2). The behavioral, device-specific and experience-related variables were measured on a five-point Likert scale (highest score being 1 and lowest score being 5). Users gave high rating for aspects such as agreeableness, excitement levels, entertainment aspect etc. of living in a smart home which is depicted in the mean scores in Fig. 2(a). The result that high rating was recorded for all these experiential attributes supports the observation that the general feedback for smart home has been favorable for most participants.

Fig. 2(b) presents the overall evaluation of utilities derived from smart homes. While nearly 70% participants rated the smart home setup high on aesthetics and energy savings, only around 58% participants gave high scores for the safety dimension. The most prominent additional feature that users wished for was related to energy savings (mean=1.66), followed by physical wellbeing (mean=2.41) and emotional well-being (mean=2.5). Concerns regarding cost stood out more prominently than concerns relating to privacy (Fig. 2(c)).

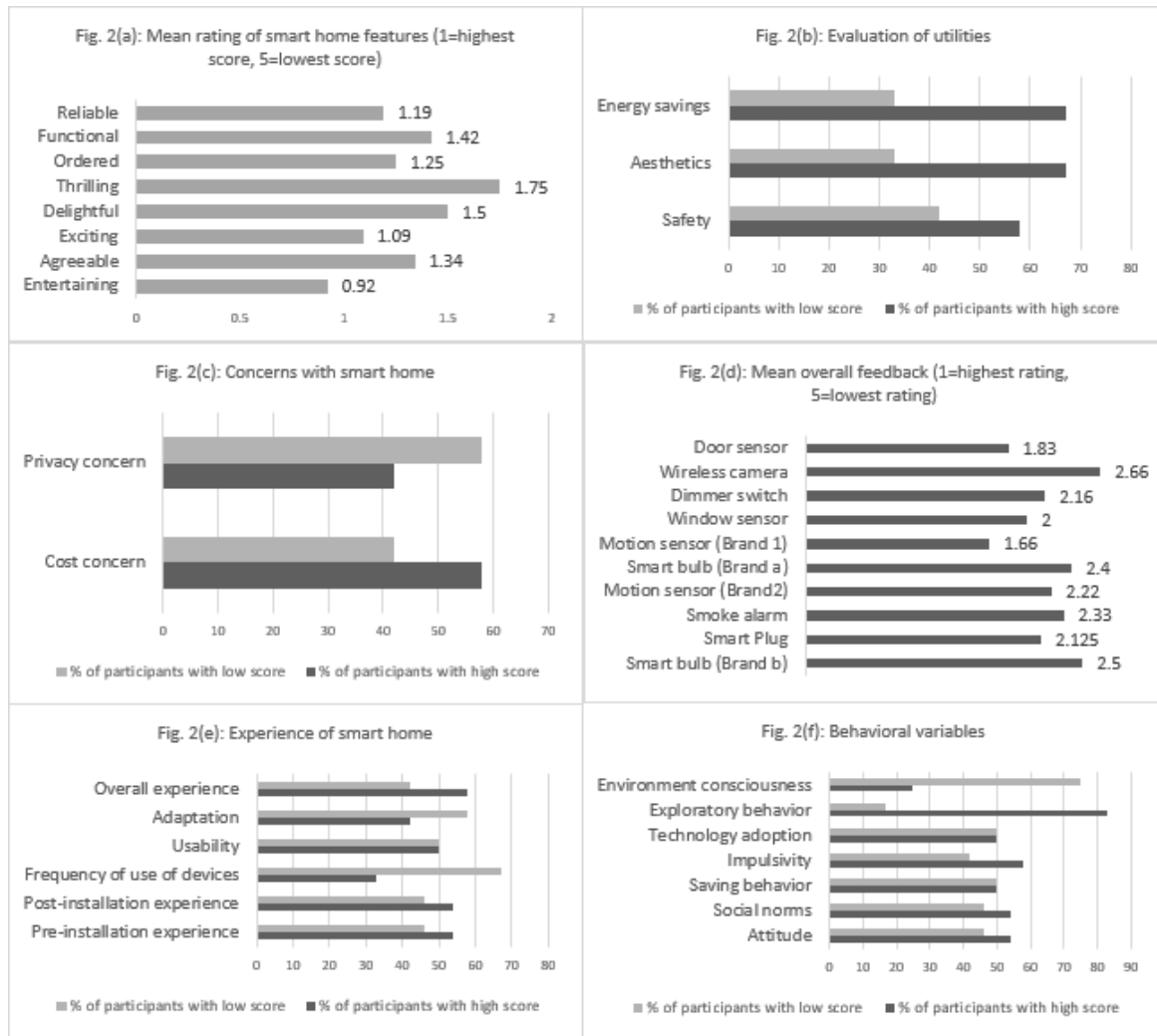


Figure 2: Behavioral underpinnings of smart home experience

The devices which were mostly installed by participants were the smart plug (83%) and one brand of smart bulbs (83%), while only 30% of the participants set up the window sensor. The maximum number of devices were set up in the living room. In terms of device-specific experience, a particular brand of the motion sensor was rated highest (mean=1.66) and the wireless camera (mean=2.66) the lowest (Fig. 2(d)).

Fig. 2(e) shows that most participants had an overall positive experience with smart home as evident from the proportion of high scorers for pre-installation experience (54%), post-installation experience (54%) and overall experience (58%). But the proportion of low scorers were more than high scorers for both adaptation and frequency of use of devices (58% and 57% registered low scores respectively). Thus, it could be inferred that while pre-installation and post-installation experience were positive (rendering an overall positive experience), the installation and use of devices phase were met with less than ideal response. The reasons for this are

discussed in the sections that follow. Fig. 2(f) shows that for the all variables except environment consciousness (Individual's willingness to adopt environment friendly measures), the percentage of high scorers exceed (or are at least equal to) that of low scorers. Notably while only 25% of respondents scored high on environment consciousness, 83% scored high on exploratory behavior (measure of individual's initiative in finding out more about their surroundings).

4.3 User personas

Apart from their experience from receiving the kit to post-installation, we also coded the qualitative descriptions of experience of living in a smart home that emerged in the interviews to arrive at personas. Two researchers analyzed the qualitative data to arrive at codes which were clubbed into four key themes- namely convenience, privacy, safety and energy savings. The participants were then categorized based on the major need which they expressed. This was then corroborated with data from the questionnaires by a

third researcher. Based on this exercise, four major personas were identified.

4.3.1 Convenience-oriented. Convenience is used to refer to the way smart devices may have aided in making activities at home easier or more enjoyable. All the participants mentioned at least one way in which smart devices had made life more comfortable for them, with some highlighting a more pronounced effect than others. From the questionnaires, the percentage of people who highlighted convenience more prominently than other aspects such as safety or sustainability is 50%. One of the major aspects which respondents brought up was a sense of control over their home, where they could sit at any place and regulate the devices. The smart home setup was said to be helpful in reducing the cognitive load of dealing with devices. The devices helped them reduce stress, have uninterrupted sleep, think less about tasks at home and make easier the tasks they were lazy to do. As one of the users pointed out, ‘users become smarter in a smart home’ by being able to manage tasks better. For others, the setup was a source of cognitive engagement, through having fun setting up, trying out different combinations and reading up about smart devices. Reference was also made to how the ambience of the home improved through the sleek design of the devices as well as mood lighting. Although some participants mentioned that adjusting to the setup was difficult because of ‘muscle memory’ which made them go to the switch board by habit, the overarching feelings associated with the setup was positive, with many conversations being peppered with descriptors such as ‘exciting’, ‘happy’, ‘cozy’, ‘fun’ and ‘easy’. Behavioral characteristics for this persona included having a positive attitude towards technology, belief that technology works for the good of people, curiosity towards trying new gadgets, and eagerness to embrace adoption of smart devices in their lifestyle. Voice command features and better interactivity within devices came in as suggestions from this cohort to further enhance the functioning of smart homes.

4.3.2 Safety-oriented. More than half of the participants also mentioned how the smart devices helped them in realizing their need for safety and security at home. 20% participants ranked this aspect as the most important utility of a smart home. While some mentioned safety as their most fundamental need right at the beginning of the interview, many placed more weightage to this need after engagement with certain smart devices. When talking about the safety aspect, participants brought up incidents of burglaries in their neighborhood. Others mentioned how without smart devices, they would have to rely on neighbors to check up on their house when they were away. The camera was said to have been useful in monitoring activity inside and outside the house as well as to check visitors before letting them enter the house. The smart bulbs were said to have been switched on in the evenings when no one would be at home, so that anyone attempting a break-in would think the house was occupied. Those looking out for safety features as a major takeaway had independent houses, belonged to an older age group and were living with kids in the family. Motion sensors, window sensors and cameras in general seem to be devices most liked by this group.

4.3.3 Cautiously supportive. Although privacy as an aspect was an explicit question in the interview schedule, many participants

brought it up on their own before we asked them about it. 58% of the participants evaluated privacy as a major concern. However, a quarter of the respondents in the interview phase mentioned that they had no concerns of privacy or data leakage when using the devices. While some mentioned how data capture was part of the digital era and hence something which they did not bother about, others mentioned that they trusted the service providers to not misuse the data. But most participants mentioned that the data captured and how it was being used were of interest to them, especially with regard to the camera. Half the participants mentioned that they were uncomfortable setting up the camera in the private space of the home and expressed concerns such as ‘data being sent somewhere’ or ‘somebody accessing videos and images’. The participants who had high concerns about privacy belonged to the older aged cohort, were mostly living with families and kids, had apprehensions about technology and experienced the feeling of ‘being recorded’.

4.3.4 Sustainability-oriented. More than half of the participants expressed interest in smart devices specifically for their sustainability features. For 30% participants, this aspect emerged as the most important utility of a smart home relative to convenience or safety. They believed that smart devices, especially the smart bulbs, plug and motion sensor helped them reduce their energy use. Many were interested in obtaining data specifically relating to monthly energy savings. People who explicitly expressed an inclination for eco-friendly devices and lifestyles in general were more insistent on this demand. This persona is keen on energy conservation, are in their early 30s, look for specific inputs from smart homes on inputs on energy conservation and have taken specific measures in terms of energy conservation while installing devices, such as placing a motion sensor and bulb together to reduce energy consumption. Additional recommendations also flowed in from this persona, such as integrating feedback on physical activities and fitness levels or recording sleep patterns.

5 DISCUSSION AND CONCLUSION

This study is aimed at understanding the finer behavioral underpinnings of the user’s interactions with a smart home. A paradigm shift in the adoption of smart home technologies is only possible through increased levels of awareness among customers and the ability of the service providers to gauge the needs of the customers. The interaction of various devices is seen as an important determinant in providing a comfortable ambience. Greater accessibility and adoption rates are possible with improvements made towards simplifying the setting up process, thus enabling an enriched user experience. Towards this, the study finds that a clear majority of the participants considered the transition to a smart home as positive, albeit some hiccups during the installation process. Nevertheless, the participants are quite optimistic about the ability of smart homes to contribute significantly to their lifestyle. The four user personas observed in this study could be factored in by service providers while designing their offerings and services, thus paving way for adequate customization to diverse user needs. Suggestions for combinations of smart devices to target a particular persona could be made since users would find it easier to connect with this larger application than specific utilities of individual devices. A limitation of

the study is that it has not been tested out in a sample that includes persons who do not identify as tech-savvy or are not acquainted with gadgets in general. Also, quantitative data-driven personas require a larger sample size which the study lacks. Future work in this space could be expanded with analysis of data from sensor devices' logs to observe patterns of installation and use of devices. This data could be triangulated to give customized smart home solutions by suggesting real-time setups and features that cater to specific needs of users.

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